



T1: Create Submission

Core Service

COMPENSATABLE



T2: Execute SQL

Judge Service

PIVOT



T3: Update Score

Core Service

RETRIABLE



T4: Notify Student

Notification

RETRIABLE



STEP

TYPE

COMPENSATION

T1 Create Submission

COMPENSATABLE

→ Mark ERROR

T2 Execute SQL

PIVOT

— (go/no-go)

T3 Update Score

RETRIABLE

— (must succeed)

T4 Notify Student

RETRIABLE

— (must succeed)



- Click a button to start the saga



Trong KBLab LMS: Hệ thống sử dụng implicit saga (choreography via Kafka). Submit flow gồm 3-4 steps: Core Service tạo submission → Kafka → Judge Service chấm SQL → Kafka → Core Service cập nhật score → WebSocket notification. Khi Judge gặp lỗi, compensation = đổi status thành ERROR + thông báo "vui lòng thử lại".